

# PalmQ10

## Cluster Connection Manual

Version 0.1.3-alpha

Running simulations on HPC

SLURM and SGE schedulers • Lustre storage

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## 1 Overview

PalmQ10 can send the computationally heavy stages of a simulation to an HPC cluster instead of running them on your workstation. This is worthwhile when your domain is large, your period is long, or your resolution is fine enough that a local run would be impractical.

### 1.1 What runs where

| Stage                                    | Runs on        | Notes                                               |
|------------------------------------------|----------------|-----------------------------------------------------|
| Prepare / create static                  | Local          | Downloads source data and builds the static driver. |
| Prepare dynamic                          | Local          | Writes configuration files.                         |
| Create dynamic<br>(WRF/WPS,<br>WRF4PALM) | <b>Cluster</b> | Usually the longest stage.                          |
| Run PALM                                 | <b>Cluster</b> | The simulation itself.                              |
| Post-processing                          | Local          | Produces maps and slices from downloaded output.    |

PalmQ10 handles the transfer in both directions: it uploads inputs, submits the job to the scheduler, monitors it, and downloads results when it finishes.

### 1.2 What you need

- An account on a cluster running **SLURM** or **SGE**, with Lustre or similar shared storage.
- Apptainer available on the cluster (or the ability to run the supplied images).
- Enough scratch quota for the container images (about 4.6 GB) and the geographic dataset (about 28 GB extracted).
- Your cluster login password, for one-time setup only.

## 2 How authentication works

Understanding this saves confusion later, and it is the largest change in this release.

### 2.1 The problem it solves

Previously, every stage of a cluster job authenticated separately. A single run prompted for your cluster password four or five times, and because university clusters are typically reached over a VPN that drops or times out, any interruption forced yet another prompt. Long unattended transfers were effectively impossible.

### 2.2 The two passwords

These are different things and it matters that you keep them straight.

| Password                      | What it is                                                                                                                                                                                                           |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cluster login password</b> | The password your HPC centre issued you. You type it <b>once, ever</b> — during first-time setup. PalmQ10 uses it a single time to register its key with the cluster, then discards it. It is never written to disk. |
| <b>Master password</b>        | A password <b>you choose</b> , local to your computer. It unlocks the encrypted vault holding PalmQ10's login key. You type it <b>once per job</b> .                                                                 |

## 2.3 What happens during a job

You enter your master password once. PalmQ10 unlocks the vault, loads its key into a temporary agent that exists only for that job, and runs everything — uploads, submission, monitoring, downloads — without asking again. When the job ends the agent is destroyed and the unlocked key is cleared from memory.

Because login is key-based rather than password-based, a dropped VPN or WiFi connection reconnects silently. Interrupted file transfers resume where they stopped.

### Why once per job rather than once and done?

Deliberate. The key is unlocked only while a job is running, so an idle session on an unattended machine cannot be used to reach the cluster.

## 2.4 Security properties

- Your cluster login password is **never stored** by PalmQ10.
- The login key is held in a vault encrypted with your master password, using scrypt key derivation and authenticated encryption.
- Vault and key files are readable only by your user account.
- Your master password is never saved, transmitted, or displayed.
- Someone who copies your laptop's files still cannot reach the cluster without your master password.

### There is no master password recovery

This is what makes the vault worth having — a recovery mechanism would be a way around it. If you forget it, re-run cluster setup: PalmQ10 generates a fresh key and re-registers it, and you enter your cluster login password once more. Nothing on the cluster is damaged.

## 3 The cluster profile

A profile is a YAML file describing one cluster. Two are supplied, for the UPV Rigel and Sirius systems, in:

```
Documents\palmq10\documentation\cluster_profiles\
```

### The supplied profiles contain the maintainer's username

Before your first job you **must** change **username** and every path containing it to your own account. The supplied values will not work for you.

### 3.1 Fields you must edit

| Field                             | Meaning                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>username</code>             | Your cluster login name. On some systems this includes a domain, for example <code>yourname@upvnet.upv.es</code> .                                           |
| <code>remote_project_root</code>  | Where PalmQ10 keeps job files in your home directory.                                                                                                        |
| <code>remote_scratch_root</code>  | Working directory for running jobs. Should be on scratch, not home — jobs write heavily here.                                                                |
| <code>remote_geog_dir</code>      | Where the geographic dataset lives on the cluster.                                                                                                           |
| <code>remote_container_dir</code> | Where the container images live.                                                                                                                             |
| <code>remote_assets_root</code>   | Shared asset root (present on the Sirius profile).                                                                                                           |
| <code>remote_images</code>        | The three per-container paths beneath it — <code>palm</code> , <code>wrfwps</code> , <code>wrf4palm</code> . Each contains the username and must be updated. |

#### Changing username alone is not enough

The username is embedded in every remote path above, so editing only the `username` line leaves the profile pointing at another user's home and scratch directories. The job then fails with permission errors that give no obvious hint that the profile is at fault. Search the whole file for the old username and replace every occurrence.

### 3.2 Fields describing the cluster

| Field                          | Meaning                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>cluster_name</code>      | Identifier used in logs and the interface.                                                                            |
| <code>login_host</code>        | Hostname you SSH into.                                                                                                |
| <code>ssh_port</code>          | Usually 22.                                                                                                           |
| <code>scheduler</code>         | <code>slurm</code> or <code>sge</code> .                                                                              |
| <code>storage_type</code>      | <code>lustre</code> or <code>scratch_lustre</code> .                                                                  |
| <code>container_runtime</code> | <code>apptainer</code> .                                                                                              |
| <code>auth_mode</code>         | <code>key_vault</code> for one master password per job, or <code>password_interactive</code> for the older behaviour. |

### 3.3 Scheduler-specific fields

SLURM and SGE describe resources differently, so the relevant fields differ.

|               | SLURM (e.g. Sirius)                                                                                 | SGE (e.g. Rigel)                                                  |
|---------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Work grouping | <code>default_partition</code> ,<br><code>default_qos</code>                                        | <code>default_queue</code> ,<br><code>default_parallel_env</code> |
| Task layout   | <code>default_ntasks</code> ,<br><code>default_cpus_per_task</code> ,<br><code>default_nodes</code> | derived from slots                                                |
| Memory        | <code>default_memory_total</code>                                                                   | <code>default_memory_per_core</code>                              |

### 3.4 Resource limits

These prevent PalmQ10 requesting more than a node can provide.

| Field                                   | Meaning                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <code>max_single_node_slots</code>      | Cores available on one node.                                                             |
| <code>node_memory_gb</code>             | Physical memory per node.                                                                |
| <code>memory_safety_fraction</code>     | Fraction of node memory PalmQ10 will request, leaving headroom for the operating system. |
| <code>minimum_memory_per_core_gb</code> | Floor below which requests are not reduced.                                              |
| <code>wrf4palm_mem_per_core_gb</code>   | WRF4PALM's memory requirement, which is much higher than the other stages.               |
| <code>default_walltime</code>           | Job time limit. Raise for long simulations.                                              |

#### Walltime is a common cause of lost work

The supplied profiles request **50 hours** on Rigel and **60 hours** on Sirius. These values come from real production runs and are deliberately generous — a job killed at its time limit loses everything computed so far. If you shorten them, be confident your run finishes inside the limit. Note also that your centre's queue policy may cap walltime below what the profile requests, in which case the queue wins.

## 4 First-time setup

### 4.1 Prepare WSL

The cluster steps run inside WSL. Confirm the required tools are present:

```
sudo apt update && sudo apt install -y rsync openssh-client
```

The vault also needs the `cryptography` Python package in WSL's `python3`, which is usually already installed. If setup reports it missing:

```
python3 -m pip install --user cryptography
```

#### You do not need rsync or an SSH client on the Windows side

PalmQ10 hands cluster work to WSL automatically. If `where.exe rsync` finds nothing on Windows, that is expected and not a problem.

### 4.2 Edit your profile

Open the profile for your cluster and set `username` and the paths that contain it. Confirm `auth_mode` is `key_vault`.

### 4.3 Create your master password

In the cluster profile dialog, choose **Set / change HPC master password**. You will be asked to type a new password twice.

Choose something you will remember — there is no recovery. The master password is **shared across all clusters** on your machine, not one per cluster.

## 4.4 Register your key

On your first cluster operation, PalmQ10 creates a login key and registers it with the cluster. You will be asked for your **cluster login password once** during this step. PalmQ10 uses it to install the key, verifies that key login works, and discards the password.

From this point you will not be asked for your cluster password again.

### Adding a second cluster later

You reuse the same master password. You will enter that new cluster's login password once during its setup.

## 5 Uploading cluster assets

Before your first job, the container images and geographic dataset must exist on the cluster. This is a one-time operation per cluster.

### 5.1 Check what is already there

Click **Check HPC assets**. PalmQ10 reports which images and datasets are present and which are missing.

### 5.2 Upload what is missing

Separate buttons upload the PALM, WRF/WPS, and WRF4PALM images, and prepare the WPS\_GEOG dataset.

### These transfers are large

The container images total roughly 4.6 GB; the geographic dataset is larger still. Start them when you can leave the machine running — overnight is sensible.

Asset uploads retry **indefinitely** rather than giving up. Combined with key-based authentication, this means a transfer survives a VPN session that times out overnight: the connection re-establishes silently and the transfer resumes where it stopped. Progress is reported as it goes, and you can interrupt with Ctrl-C.

### 5.3 A note on the geographic dataset

The WPS\_GEOG upload also applies a compatibility fix. One dataset folder is named `con`, which is a reserved device name on Windows and therefore cannot exist as a folder there. PalmQ10 stores it locally under a safe name and restores the correct name on the Linux side during upload. This is automatic.

## 6 Running a job

1. Set **Computing backend** to HPC.
2. Select your cluster profile.
3. Set **Available cores** to match what your allocation provides.
4. Run the stages as normal.

When a cluster stage begins you are asked for your master password once. After that the job proceeds unattended: inputs upload, the job is submitted, PalmQ10 polls the scheduler, and results download on completion.

**Keep the session open**

PalmQ10 monitors the job from your machine. Closing the dialog or shutting down stops monitoring and the automatic download, though the job itself continues on the cluster.

## 7 Adding a new cluster

There is no per-cluster special handling in the code. To add a cluster:

1. Copy an existing profile — start from the SLURM one for a SLURM cluster, the SGE one for SGE.
2. Set `cluster_name`, `login_host`, and `username`.
3. Set the remote paths for your account, putting scratch paths on scratch storage.
4. Set `scheduler` and `storage_type`.
5. Set the queue or partition names your centre uses.
6. Set `max_single_node_slots` and `node_memory_gb` from your cluster's node specification.
7. Leave `auth_mode: key_vault`.
8. Run **Check HPC assets** and upload whatever is missing.

Your centre's documentation, or `sinfo` on SLURM and `qconf -sql` on SGE, will tell you the queue and node details.

## 8 Troubleshooting

| Symptom                                | Cause and resolution                                                                                                                                                    |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Asked for the cluster password mid-job | Key authentication is not active. Usually a first-time setup that did not complete, or <code>auth_mode</code> not set to <code>key_vault</code> . Re-run cluster setup. |
| Asked for a key passphrase repeatedly  | The vault was not unlocked before that operation began. Report it — this indicates a code path that is not acquiring the session up front.                              |
| Master password rejected               | Three attempts are allowed. If you have genuinely forgotten it, re-run cluster setup to generate a fresh key.                                                           |
| Job submitted then vanished            | Usually walltime exceeded. Check the scheduler log on the cluster and raise <code>default_walltime</code> .                                                             |
| Out-of-memory on WRF4PALM              | WRF4PALM needs far more memory per core than the other stages. Check <code>wrf4palm_mem_per_core_gb</code> and consider a smaller domain.                               |
| Geogrid fails immediately              | Usually an incomplete geographic dataset — a dataset folder missing its <code>index</code> file. Re-run the WPS_GEOG upload.                                            |



| Symptom                          | Cause and resolution                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Transfers stall and never resume | Confirm you can reach the login host from WSL at all. If the VPN is fully down, PalmQ10 will keep retrying and resume once it returns. |

## 8.1 Where to look on the cluster

Job working directories are under your `remote_scratch_root`. By default PalmQ10 does not delete them after a successful run (`cleanup_remote_on_success: false`), so scheduler logs and model output remain available for inspection. Clear them periodically — they consume scratch quota.

## 9 Further documentation

- *User Guide* — the interface and a worked example.
- *Installer Guide* — installation and first-use setup.
- *Technical Guide* — architecture and internals, for developers.